# SOFTWARE REQUIREMENTS SPECIFICATION

## ImpactCocoa Data Management Platform

Version 4 | December 2025

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| **Document Title** | Software Requirements Specification (SRS) |
| **Version** | Version 4 — December 2025 |
| **Prepared by** | Think! Data Services Ltd. |
| **Client** | AgroEco Ghana — Werkmaatschappij Agro Eco |
| **Development Partner** | Ondora Global JSC |
| **Contract Reference** | TDS-AG-2026-001 |
| **Project Start** | 24 March 2026 |
| **Target Go-Live** | 21 July 2026 (16 weeks) |
| **Document Status** | Final — forms part of the signed contract as Schedule A |

**Note:** *This document is an extract of Schedule A from the signed software development agreement between Think! Data Services Ltd. and AgroEco Ghana (TDS-AG-2026-001). It is provided to Ondora Global JSC for development purposes only. Commercial terms, payment schedules, and bank details are not included. Do not distribute beyond the assigned development team.*

# 1. INTRODUCTION

## 1.1 Purpose

This Software Requirements Specification (SRS) provides a comprehensive description of the ImpactCocoa Data Management Platform to be developed by Ondora Global JSC on behalf of Think! Data Services Ltd. for AgroEco Ghana.

The platform aims to enhance field operations, reporting, and data analytics by collecting, storing, processing, and analysing farmer and cocoa supply chain data in compliance with Rainforest Alliance (RA) Cocoa Standard and EU Deforestation Regulation (EUDR) requirements.

This SRS is the foundation for all system design and development phases. Every deliverable will be validated against the requirements set out in this document at each project gate.

## 1.2 Project Scope

AgroEco Ghana provides services to four cocoa farmer cooperatives and seeks to digitise and integrate its core operational functions. The platform will cover:

- Approximately 4,000 registered cocoa farmers across 4 cooperatives
- Cooperative locations: Assin Fosu, Asankrangwa, THV, and Daboase
- Data collection via KoboToolbox with offline-first capability
- EUDR compliance — semi-automated by design to control development cost; fully automated assessment can be added in a later phase
- Rainforest Alliance Cocoa Standard certification tracking and reporting
- Migration of 4 years of historical data from the current FarmForce system

## 1.3 Key Modules

- Farmer Profiling — registration, household data, certification status, field pre-assessments
- Farm Inspections — annual compliance status, RA inspection details, non-compliance follow-up
- Farm Mapping — GPS/GIS farm field boundaries, EUDR deforestation status
- Traceability — cocoa bean tracking from farm gate to point of export
- Farmer Training & Coaching — training sessions, attendance, coaching visits, Farm Development Plans
- Reporting & Analytics — dashboards, RA reports, EUDR compliance, traceability reports
- System Administration — user management, RBAC, audit logs

**Note:** *EUDR assessment is semi-automated. Farm boundaries are checked against a reference deforestation dataset via QGIS. The database stores and displays the result. Full automation of the assessment process is out of scope for this version and can be added in a subsequent phase.*

# 2. OVERALL DESCRIPTION

## 2.1 Product Perspective

The ImpactCocoa Data Management Platform is a new standalone cloud-based system developed to address challenges around data management at the cooperative level, with a focus on traceability, Rainforest Alliance Cocoa Standard compliance, and EUDR requirements. It replaces the current FarmForce system.

The system operates with a client-server architecture supporting both online and offline data collection through mobile applications, with automatic synchronisation when connectivity is restored. It integrates with KoboToolbox for field data collection and exposes RESTful APIs for third-party integration.

## 2.2 User Classes, Characteristics, and Data Scope

The following user classes must be implemented with distinct permissions. District-level data segregation is a core architectural requirement — it must be enforced at the API layer, not only at the UI layer.

| User Class | Description | Primary Needs / Modules | Data Scope |
|---|---|---|---|
| **Field Officers** | Primary contact with farmers; collect field data | Farmer Profiling, Farm Mapping, Farm Inspections, Farmer Training, Coaching. Selected Field Officers also handle Traceability. | **Own district only** |
| **IMS Managers** | Handle data collection, cleaning, validation; organise training | All modules, Data Validation, Report Generation | **Own district only** |
| **Project Leader** | Oversees all operational activities; final RA validation layer | Analytics, Dashboard, All Reports | **All districts** |
| **System Administrator** | Responsible for administrative tasks | User Management, System Configuration, Report Access | **All districts** |
| **Whittakers** | External viewer | Analytics, Dashboard, Reports — View Only | **All districts** |
| **Cooperatives (Chair)** | Cooperative President/Chairman | Analytics, Dashboard, Reports — View Only | **Own district only** |

**Note:** *Data segregation is non-negotiable. Field Officers and IMS Managers must never see data outside their assigned cooperative/district. Only Whittakers and the Project Leader have cross-cooperative visibility. All data-modifying API endpoints must validate user permissions against cooperative assignment before processing.*

## 2.3 Product Functions Summary

The platform provides the following major functional areas:

- User Authentication — login, password reset, JWT session management, RBAC
- Farmer Profiling — profile creation and editing, photograph upload, household members
- Farm Inspections — annual inspection records, compliance status, non-compliance follow-up
- Farm Mapping — GPS polygon capture, field ID generation, EUDR overlay, Leaflet viewer
- Traceability — purchase records (primary and secondary evacuation), batch numbering
- Farmer Training & Coaching — training modules, attendance, coaching visits, Farm Development Plans

- Reporting & Analytics — RA, EUDR, traceability, and training reports; dashboard visualisations
- System Administration — user management, role/permission management, audit logs

# 3. FUNCTIONAL REQUIREMENTS

All requirements in this section are identified with a unique ID (FR-XX) for traceability. Every requirement must be demonstrated at the relevant project gate before acceptance is granted.

## 3.1 User Authentication

| ID | Requirement |
|---|---|
| **FR-1** | The system shall provide a login screen through which users access the system |
| **FR-2** | The system shall authenticate users through their email address and password |
| **FR-3** | The system shall allow users to reset forgotten passwords |
| **FR-4** | The system shall send a reset link for forgotten passwords and require users to set a new password |
| **FR-5** | The system shall implement role-based access control (RBAC) to restrict features and data based on user role and cooperative assignment |

## 3.2 Farmer Profiling

| ID | Requirement |
|---|---|
| **FR-6** | The system shall extract profiling data of farmers from the annual internal inspections form, with fields for personal information, contact details, and identification |
| **FR-7** | The system shall automatically generate a unique farmer code for each farmer profile |
| **FR-8** | The system shall allow editing of farmer profile information |
| **FR-10** | The system shall allow uploading of farmer photographs where available |
| **FR-11** | The system shall record personal information on each household member of a farmer |
| **FR-12** | The system shall support bulk import of farmer data from external sources including CSV files and KoboToolbox |
| **FR-13** | The system shall maintain a complete audit trail of all changes made to farmer profiles, recording old values, new values, user ID, and timestamp |

## 3.3 Farm Inspections

| ID | Requirement |
|---|---|
| **FR-14** | The system shall import a selection of data for recording annual inspection details including inspection date, inspector name, and findings |
| **FR-15** | The system shall allow updating of farmer compliance status based on inspection results |
| **FR-16** | The system shall allow assigning and managing follow-up actions based on farmer non-compliances; the database shall include follow-up actions and due dates |
| **FR-17** | The system shall calculate the farmer score and certification status automatically |
| **FR-18** | The system shall maintain a complete inspection history for each farmer with timestamps and inspector details |

## 3.4 Farm Mapping

| ID | Requirement |
|---|---|
| **FR-19** | The system shall allow linking of farm parcels to farmer profiles |
| **FR-20** | The system shall automatically generate a unique field ID for each farm parcel |
| **FR-21** | The system shall capture and store GPS coordinates and polygon data for farm boundaries (PostGIS geometry, SRID 4326). Parcels outside Ghana bounding box (4.5–11.5°N, 3.5°W–1.5°E) shall be rejected. Parcels within 10 metres of an existing parcel shall be flagged. |
| **FR-22** | The system shall import geospatial data from external sources including KML, KMZ, Shapefile, and GeoJSON formats |
| **FR-23** | The system shall record field characteristics including size, crop type, planting date, and tree count |
| **FR-24** | QGIS shall be deployed to analyse deforestation status per EUDR requirements by comparing farm boundaries against forest baseline data. This is a semi-automated process. |
| **FR-25** | The system shall display visual farm maps with parcel boundaries on satellite imagery and show the deforestation status per field via an interactive Leaflet.js viewer |
| **FR-26** | The system shall calculate farm area automatically based on GPS polygon data |
| **FR-27** | The system shall export geospatial data in standard formats (GeoJSON, KML, Shapefile) for use in external GIS applications |

## 3.5 Traceability

| ID | Requirement |
|---|---|
| **FR-28** | The system shall record cocoa purchases at primary and secondary evacuation level, including farmer ID, quantity bought, quality grade, and purchase date |
| **FR-29** | The system shall register unique batch numbers for cocoa purchases to enable tracking from farm gate to export |
| **FR-30** | The system shall generate complete traceability reports showing the journey from farmer to export for EUDR compliance: Batch ID → Purchases → Parcels → GPS polygons |

## 3.6 Farmer Training & Coaching

| ID | Requirement |
|---|---|
| **FR-31** | The system shall allow creation of training modules with title, description, and objectives |
| **FR-32** | The system shall allow assignment of farmers to training sessions |
| **FR-33** | The system shall record training attendance |
| **FR-34** | The system shall register training coaching visits — date, attendees, and actions agreed upon |
| **FR-35** | Coaches shall be able to collect data for the Farm Development Plan |
| **FR-36** | Coaches shall have access to follow-up actions of farmers for effective coaching sessions |
| **FR-37** | Coaches shall be able to create individual coaching field reports to track farm and farmer improvement |

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|---|---|
| **FR-38** | The system shall generate training reports showing participation rates |

## 3.7 Reporting & Analytics

| ID | Requirement |
|---|---|
| **FR-39** | The system shall include reporting templates for the following: Primary Evacuation report, Secondary Evacuation report, Traceability report, Internal Inspection and follow-up actions report, Training and Coaching report, GRM report |
| **FR-40** | Reports shall generate within acceptable timeframes based on data volume (see NFR-4) |
| **FR-41** | The system shall allow users to generate reports with customisable parameters (date range, cooperative, farmer, season) |
| **FR-42** | The system shall support multiple export formats for all reports (PDF, Excel, CSV) |
| **FR-43** | The system shall enforce role-based access control for reports — users only see reports for their own district/cooperative unless authorised for all districts |
| **FR-44** | The system shall maintain an audit log of all report generation activities |

# 4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements will be verified during Gate 4 (QA & Testing). All items below are contractually binding.

## 4.1 Performance

| ID | Requirement |
|---|---|
| **NFR-1** | The system shall support at least 100 concurrent users without performance degradation |
| **NFR-2** | The system shall load pages within 3 seconds under normal network conditions |
| **NFR-3** | The system shall process bulk data imports of up to 10,000 records within 5 minutes |
| **NFR-4** | Standard reports shall be generated within 30 seconds |

## 4.2 Security

| ID | Requirement |
|---|---|
| **NFR-5** | The system shall encrypt all data in transit using TLS 1.3 or higher |
| **NFR-6** | The system shall encrypt all sensitive data at rest using AES-256 encryption |
| **NFR-7** | User passwords shall be hashed using bcrypt with a minimum cost factor of 12 |
| **NFR-8** | The system shall implement role-based access control (RBAC) with granular permissions enforced at the API layer |
| **NFR-9** | The system shall log all user activities and maintain a full audit trail for compliance purposes |
| **NFR-10** | User sessions shall automatically expire after 30 minutes of inactivity |

## 4.3 Availability & Reliability

| ID | Requirement |
|---|---|
| **NFR-11** | The system shall maintain 99.5% uptime during business hours |
| **NFR-12** | Daily automated backups shall be configured; restoration on best-effort basis |
| **NFR-13** | The system shall have a recovery time objective (RTO) of 4 hours in case of system failure |

## 4.4 Usability

| ID | Requirement |
|---|---|
| **NFR-14** | The system shall have an intuitive interface requiring minimal training for basic operations |
| **NFR-15** | The system shall be responsive and accessible on desktop, tablet, and mobile devices; all screens usable on Android and iOS without horizontal scroll |
| **NFR-16** | The system shall provide clear error messages and validation feedback |

| | |
|---|---|
| **NFR-17** | The system interface shall support English language with provisions for future multilingual support |

## 4.5 Scalability

| ID | Requirement |
|---|---|
| **NFR-18** | The system shall be designed to accommodate growth to 10,000 farmer profiles |
| **NFR-19** | The database shall be designed to handle 1 million traceability records |
| **NFR-20** | The system architecture shall support horizontal scaling to accommodate increased load |

# 5. KOBOTOOLS FORM SPECIFICATIONS (Schedule D)

Eleven KoboToolbox forms are in scope. All forms must be deployed on KoboToolbox Premium, configured for offline-first use via KoboCollect Android, and integrated with the ImpactCocoa database via the Python REST sync service.

Gate 2 acceptance requires all 11 forms to be published, tested in offline mode with a minimum of 20 test submissions per form, and confirmed syncing to the staging database. Field officer training for at least one cooperative (minimum 4 participants, sign-off sheet delivered) must also be completed before Gate 2 can be accepted.

| Ref | Code | Form Name | Completed By |
|---|---|---|---|
| **F1** | IMP-INSP | Internal Inspection Form — full RA checklist, skip logic, severity ratings | *Field Officer / IMS staff* |
| **F2** | IMP-PRIM | Cocoa Purchase Form — primary evacuation; field linkage, weight/price, season tag | *Purchasing Clerk / Field Officer* |
| **F3** | IMP-SEC | Cocoa Purchase Form — secondary evacuation | *LBC / Field Officer* |
| **F4** | IMP-QC | Quality Control Form | *Depot Officer / IMS staff* |
| **F5** | IMP-SHADE | Shade Trees Monitoring Form | *Field Officer* |
| **F6** | IMP-CLMRS | CLMRS Monitoring Form | *Child Labour Officer / CDO* |
| **F7** | IMP-TRAIN | Farmer Training Record | *Field Officer / Trainer* |
| **F8** | IMP-COACH | Farmer Coaching Visit Form | *Field Officer / Coach (CDO)* |
| **F9** | IMP-VSLA | VSLA Form | *Field Officer / IMS staff* |
| **F10** | IMP-YIELD | Yield Estimation Form | *Field Officer / IMS staff* |
| **F11** | IMP-FARMER | Farmer Entrance Form — GPS capture, all profiling fields, validation rules | *Field Officer / IMS staff* |

# 6. SYSTEM ANALYSIS

## 6.1 Technology Stack

The following technology stack is agreed with the client and must not be deviated from without written approval from Think! Data Services Ltd.

| Component | Technology |
|---|---|
| **Frontend Framework** | React.js — responsive web interface, mobile-first |
| **Backend Framework** | Node.js with Express |
| **Database** | PostgreSQL with PostGIS extension for geospatial data (SRID 4326) |
| **Cloud Hosting** | AWS or GCP — SSL, automated backups, uptime monitoring |
| **Mobile Data Collection** | KoboCollect Android app (offline-first) via KoboToolbox Premium |
| **Mapping / GIS** | Leaflet.js for interactive web maps; QGIS for desktop deforestation analysis |
| **Reporting** | Custom dashboards with Chart.js |
| **Authentication** | JWT / OAuth 2.0 — role-based session management |
| **Kobo API Sync** | Python REST service — incremental sync with pagination and error recovery |
| **Security** | TLS 1.3 in transit; AES-256 at rest; bcrypt passwords (min cost factor 12) |
| **Integration** | KoboToolbox REST API; RESTful APIs for third-party data exchange |

## 6.2 Integration Points

- KoboToolbox REST API — offline field data collection and synchronisation via Python sync service; incremental sync with pagination and error recovery
- Third-party export — RESTful APIs for data exchange with external systems (e.g. EUDR reference datasets, RA certification systems)
- QGIS — desktop tool for deforestation analysis; outputs imported into the platform as polygon-level EUDR status flags
- FarmForce — legacy system; 4 years of historical data to be migrated via ETL process with a reconciliation report delivered to the client

## 6.3 Data Volumes (design targets)

- Farmer profiles: 4,000+ current; designed for 10,000 (NFR-18)
- Farm parcels: 10,000+; GPS polygon storage in PostGIS
- Cocoa purchase records: ~30,000/season
- Inspection records: ~12,000/season
- Traceability records: designed for 1 million (NFR-19)
- Kobo sync load test: 50,000-record full-season simulation must pass without errors (Gate 4)
- Concurrent users: minimum 100 without degradation (NFR-1)

# 7. DEVELOPMENT EFFORT & TIMELINE

## 7.1 Team & Effort Breakdown

The project requires 120 total person-days across three specialised roles working in parallel over 16 weeks.

| Role | Days | Scope |
|---|---|---|
| **Full-Stack Developer** | 82 days | All backend APIs, auth, RBAC, traceability, frontend, reporting, migrations |
| **GIS Developer** | 13 days | PostGIS setup, farm mapping, Leaflet viewer, EUDR overlay, geospatial import/export |
| **Kobo / Data Expert** | 20 days | All 11 KoboToolbox forms, offline config, API sync service, field testing, documentation |
| **Data Migration** | 5 days | FarmForce legacy ETL (4 years, 4 cooperatives), reconciliation report |

## 7.2 Phase & Timeline Overview

The project is structured into 7 phases. The timeline below is indicative. Actual dates depend on timely feedback and approvals from AgroEco Ghana. Think! Data must be notified of any risk to the agreed timeline within 5 working days of the risk being identified.

| | Weeks | Phase | Gate | Team | Key Deliverables |
|---|---|---|---|---|---|
| **P1** | Wk 1–2 | **Setup & Architecture** | **Gate 1** | *Full-Stack + GIS + Kobo* | PostgreSQL/PostGIS schema v1.0; Node.js/Express + React.js on staging; KoboToolbox workspace activated; task board live |
| **P2** | Wk 2–6 | **KoboToolbox Forms** | **Gate 2** | *Kobo Expert + Full-Stack* | All 11 forms published, tested offline (20 submissions each), syncing to staging DB; field officer training Coop 1 complete |
| **P3** | Wk 3–7 | **GIS & Geospatial Mapping** | **Gate 2** | *GIS Developer* | Farm polygon module; GPS validation; EUDR deforestation overlay; Leaflet.js interactive map; GeoJSON batch export |
| **P4** | Wk 3–10 | **Core Backend Development** | **Gate 3** | *Full-Stack Developer* | All backend APIs; JWT auth + RBAC; farmer, farm, purchase, inspection, traceability, training modules; 80% test coverage; FarmForce ETL |
| **P5** | Wk 6–11 | **Frontend & Dashboard** | **Gate 3** | *Full-Stack Developer* | Role-based dashboards; all module UIs; EUDR traceability report; analytics dashboard; mobile-responsive |
| **P6** | Wk 11–14 | **Testing & QA** | **Gate 4** | *Full-Stack + UAT Team* | Full test suite; GPS + Kobo sync load tests; EUDR chain verification; security audit; UAT with AgroEco; all staff training |
| **P7** | Wk 14–16 | **Deployment & Handover** | **Final** | *Full-Stack + Kobo Expert* | Production deployment (AWS/GCP); full FarmForce migration; technical docs; hypercare commenced |

## 7.3 Gate Acceptance Process

At each gate the following process applies:

- Think! Data Services Ltd. notifies AgroEco Ghana in writing that a milestone is ready for review
- AgroEco Ghana has five (5) Working Days to accept or provide written notice of defects
- Defects must be remedied within seven (7) Working Days of written notice
- If AgroEco Ghana does not respond within five Working Days, the milestone is deemed accepted
- Gate 4 UAT: maximum two (2) rounds of UAT revisions are included within the project fee

**Note:** *Payment to Think! Data Services Ltd. from AgroEco Ghana is released upon gate acceptance. The developer team's work must meet all gate criteria before Think! Data can accept and notify the client. Delays caused by client non-response or late data provision do not constitute developer delays.*

# 8. POST-GO-LIVE SUPPORT & HANDOVER

## 8.1 30-Day Hypercare (Weeks 14–16)

Following production deployment, a 30-day hypercare period is included within the project scope. During hypercare:

- Respond to reported issues within 24 hours on Working Days
- Critical bugs (system down, data corruption) resolved within 48 hours
- High-priority bugs (major feature broken) resolved within 5 Working Days
- Weekly check-ins between the Think! Data PM and AgroEco Ghana's designated contact

## 8.2 Technical Handover Documentation

The following documentation must be delivered as part of the Final milestone:

- API documentation (all endpoints, parameters, response formats)
- Entity-Relationship (ER) diagram for the full database schema
- Deployment runbook (step-by-step production deployment and rollback procedures)
- Environment setup guide (local development, staging, production)
- Kobo form maintenance guide — how to modify and redeploy forms post-handover
- Open-source licence documentation for all third-party libraries used

## 8.3 Training

- Office staff training: all 4 cooperatives, web users, sign-off sheets delivered
- Field officer training: all 4 cooperatives, KoboCollect, sign-off sheets delivered
- System Admin training: production user account creation and admin console

## 8.4 Post-Hypercare Support

Ongoing support and maintenance beyond Week 16 is not included in this contract. A separate Service Level Agreement (SLA) will be negotiated between Think! Data Services Ltd. and AgroEco Ghana. Three support tiers are available (Essential / Professional / Dedicated) with monthly costs ranging from USD 100 to USD 650.